MLP / CIFAR-10 | Cosine decay to half LR and global clipping at 1.0

Clipping only

Cosine only

Cosine + clipping

Training loss (log scale)

Raw gradient ρ_t

Gradient norm

$\eta\hat{\kappa}_t$

Applied LR / base LR

Baseline raw

Treatment raw

Adam input

Completed optimizer updates

Gray = baseline, blue = treatment, green = recorded Adam input. Each panel uses its own y range.
Shaded prefix through state 512 on the same Spark. The cosine reaches half LR at state 544.